Interactions between geometry, topology and algebra via projective geometry

The main focus of my research program is to explore various aspects of locally homogenous geometric structures on manifolds. In addition to the obvious connections to geometry, this area has strong connections to both topology, via the underlying manifold and the resulting deformation spaces, and algebra via the associated holonomy representations. Below I briefly outline my contributions to these areas and discuss some ongoing and future directions of research.

1 Convex projective structures

Background and context: Hyperbolic manifolds provide a rich source of examples of locally homogenous geometric structures. They are also crucially important pieces of geometrization results in low dimensions. Mostow rigidity provides a prominent dichotomy for (finite volume) hyperbolic manifolds in which surfaces have a rich deformation theory whereas manifolds in all other dimensions are completely rigid.

Generalized cusps: The geometry of the topological ends of finite volume manifolds is well understood. Each such is a product of a Euclidean manifold and a ray. This product is geometrically meaningful, in the sense that the Euclidean manifold is covered by a hyperbolic horosphere. In [25], Cooper, Long, and Tillmann propose a generalization of these types of ends for properly convex manifolds called *generalized cusps*. Their original definition was motivated by a version of the Margulis lemma from [23] and thus only required that the fundamental group of such a cusp be virtually nilpotent. In recent work with Cooper and Leitner [12], we were able to give a classification of generalized cusps and describe concretely what these types of cusp "look like". Our classification shows that in dimension n there are $n+1$ *types* of generalized cusps (numbered 0 through n). Roughly speaking, the types interpolate between unipotent cusps (type 0) and diagonalizable cusps (type $n+1$). For each type there is a certain abelian Lie group, and the fundamental group of the generalized cusp can be realized as a lattice in this Lie group. One of these Lie groups consists of the stabilizer of a hyperbolic horosphere, and so it follows that cusps in hyperbolic manifolds comprise one of the types (type 0). In particular we are able to show that, as in the hyperbolic setting, each such cusp is finitely covered by a torus times an interval, and hence have virtually abelian fundamental group.

In a subsequent paper with Cooper and Leitner [13], I was able to give several different descriptions of the deformation space of properly convex manifolds. This enabled us to determine the global topology of the deformation space. For instance, when the cusp is diffeomorphic to a torus times an interval the deformation space is a bundle over the space of Euclidean similarity structures on the torus and the fiber is a (explicit) cone in the vector space of cubic differentials.

Constructing convex projective manifolds: In this section I will outline several constructions of convex projective manifolds, separately dealing with the closed and non-compact cases.

Non-compact case: Recent work of Cooper and Tillman [26] has shown that if a finite volume hyperbolic manifold admits a properly convex structure then the ends must be generalized cusps. A complementary question involves exploring which types of generalized cusps can arise as ends of properly convex structures on hyperbolic manifolds. In dimension 3 it is known that all 4 types can arise and furthermore, that they can be constructed as deformations of complete hyperbolic structures. Type 0 corresponds to cusps in hyperbolic manifolds. In [6] I was able to construct the first explicit examples of a 3-manifolds with a type 1 generalized cusp. In [15], Danciger, Lee, and I were able to show that infinitely many hyperbolic 3-manifolds can be deformed to properly convex manifolds with type 3 cusps. In [7] I was able to prove a similar result showing that there are infinitely many hyperbolic 3-manifolds that can be deformed to properly convex manifolds with type 2 cusps.

In higher dimensions, Marquis and I [5] have been able to show that in each dimension there are properly convex manifolds with type 1 cusps. These examples are constructed by bending hyperbolic manifolds along totally geodesic hypersurfaces. These techniques have been recently generalized by Bobb [19] providing the existence of properly convex n-manifolds with cusps of type 0 through n. This leaves only the case of type $n+1$.

Closed case: An important result regarding closed properly convex manifolds due to Benoist [17], which roughly speaking says that a closed convex projective 3-manifold can be cut along a canonical collection of "totally geodesic" tori and the resulting components will all be diffeomorphic to finite volume hyperbolic manifolds. Intuitively, this construction provides a geometric realization of the JSJ decomposition. Given this result, it is natural to wonder if a closed manifold whose JSJ pieces are all hyperbolic admits a properly convex structure. In [15] I am able to make some progress on this question by proving that there is an infinite collection of cusped hyperbolic 3-manifolds whose doubles admit a properly convex structure.

Gluing equations: One of the most important tools for constructing examples of hyperbolic 3-manifolds is Thurston's gluing equations. These equations allow one to begin with a combinatorial description of a 3-manifold in terms of an ideal triangulation and build a system of complex valued equations whose solutions. Motivated by this construction, as well other recent generalizations [18, 30], Casella and I have developed a system of gluing for projective structures on 3-manifolds [10]. Our strategy is to associate *incomplete* flags to the vertices of the triangulation. As such, our gluing equations are well suited for studying several special types of projective structures such as properly convex structures, (possibly incomplete), hyperbolic structures, Anti-de Sitter structures, and half pipe structures, where such incomplete flags arise naturally.

In order to explore our new gluing equations, Casella and I have developed a piece of software. The software is in the spirit of SnapPy [27] and Regina [22], and allows the user to input a triangulations file from either Regina or SnapPy. The software will then calculate the corresponding gluing equations. It can then use the machine learning software PyTorch [40] to attempt to numerically solve the equations. Once a solution is found the software can compute the holonomy of the correponding projective structure and draw pictures of

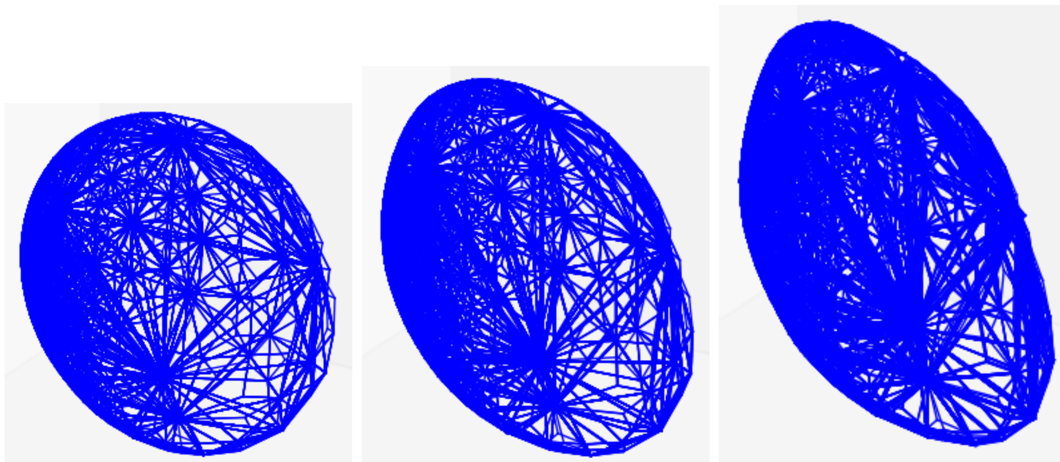

Figure 1: Developing images from solutions to gluing equations

the domain of the developing map (see Figure 1).

Future directions: Casella and I are currently exploring additional properties of our projective gluing equations. Specifically, we are attempting to determine algebraic conditions on a solution that will guarantee that the solution gives rise to a convex projective structure. Furthermore, we have recently discovered [11] some Neumann-Zagier types of relations amongst our gluing equations similar to those in [39] and [31].

I recent work with Danciger, Lee, and Marquis [14], we have shown that many Dehn fillings of the figure-8 knot complement admit non-hyperbolic convex projective structures. We believe that this phenomenon is quite general and are currently working to develop a theory of convex projective Dehn filling, and prove an analogue of Thurston's Dehn filling theorem.

2 Thin groups

Background and context: Given a semi-simple Lie group G and a lattice $\Gamma \subset G$ a group $\Delta \subset \Gamma$ is *thin* if it has infinite index in Γ and is Zariski dense in G. If one omits the infinite index hypothesis, then Δ will itself be a lattice and hence Zariski dense by the Borel density theorem. In this sense, thin groups can be thought of as infinite index cousins of lattices. These groups have seen a surge of interest in recent years due to their connection with expander graphs (see [20] for details) and affine sieving techniques [21].

There are currently many methods for constructing thin groups in a variety of Lie groups ([24, 36, 37, 28, 35, 34]), however, most of these constructions result in thin groups that are either virtually free or isomorphic to fundamental groups of surfaces.

Arithmetic hyperbolic lattices: My contribution to this area has been to provide examples of thin subgroups in a much larger set of isomorphism classes, namely fundamental groups of hyperbolic manifolds. In work with D. Long [4, 16] we are able to show that there is a large class $\mathcal{C}_n$ of arithmetic lattices in $\mathrm{SO}(n,1)$ such that for each lattice $\Gamma \in \mathcal{C}_n$ there is a finite index subgroup Γ' and infinitely many non-commensurable lattices in $\mathrm{SL}(n+1,\mathbb{R})$ each of which contains a thin subgroup isomorphic to Γ'. The class $\mathcal{C}_n$ of arithmetic lattices is explicitly describable and consists of all arithmetic lattices arising from the so called "orthogonal group" construction (see [38] for details). The class $\mathcal{C}_n$ accounts for a large portion of all arithmetic hyperbolic lattices, including all such lattices when n is even and all non-compact lattices for each n.

Non-arithmetic hyperbolic lattices: In their seminal paper [32], Gromov and Piatetski-Shapiro provided a technique for constructing infinitely many non-arithmetic hyperbolic lattices in every dimension. Their technique, which has come to be known as hybridization, involves taking two non-commensurable arithmetic manifolds, cutting them along isometric totally geodesic hypersurfaces and gluing the resulting manifolds back together. In [8] I am able to prove that up to taking a finite index subgroup, that each of the non-arithmetic lattices in $\mathrm{SO}(n,1)$ constructed in [32] can be realized as a thin subgroup of infinitely many non-commensurable lattices in $\mathrm{SL}(n+1,\mathbb{R})$.

Future directions: The previous results naturally lead to the question of which fundamental groups of hyperbolic manifolds can be realized as thin subgroups of lattices in special linear groups. In this generality, this question seems difficult to answer, in part because my previous constructions relied heavily on the deformation theory of convex projective structures, which is currently not well understood for arbitrary hyperbolic manifolds. A more tractable goal that I am currently pursuing is to prove that all arithmetic hyperbolic lattices can (virtually) realized as thin subgroups of lattices in special linear groups.

3 Complex projective structures

Background: Complex projective structures on surfaces are geometric structures locally modelled on $\mathbb{CP}^1$ with Möbius transformations as transition functions. Examples include all three constant curvature geometric structures on surface, but these structures can be far more general. Each such structure comes with a (conjugacy class of) holonomy representation. When the surface is closed, much can be said about which sorts of representations can arise as holonomies [29] and when different structures have the same holonomy [2, 3]. Roughly speaking two complex projective structures on a closed surface have the same holonomy if they are related by a sequence of geometric surgeries called 2π-*graftings*.

Non-compact surfaces: In the case of non-compact surfaces Bowers, Casella, Ruffoni, and I have made progress [9] by classifying a certain type of complex projective structures on the thrice punctured sphere. Our results deal with projective structures which are *relatively elliptic* (the holonomy of peripheral curves is elliptic) and *tame* (the developing map can be continuously extended to the punctures). Our main result is a concrete description of the moduli space of tame and relatively elliptic structures which can be roughly described as follows. There are certain "obvious" tame and relatively elliptic structures on the thrice punctured sphere that are obtained by doubling a triangle in $\mathbb{CP}^1$. Any other tame and relatively elliptic structure on the thrice punctured sphere can be obtained from one of these triangular structures by a sequence of graftings along ideal arcs (arcs connecting punctures). We are also able to show that the conditions of tameness and relative ellipticity can be described in simple terms using the language of meromorphic projective structures of Allegretti and Bridgeland [1]. Specifically, the corresponding meromorphic quadratic differential will have all poles of order 2 and exponents that are real, but not integral. This result should be thought of as complementary to recent work of Gupta and Mj [33] where they deal with meromorphic projective structures with poles of higher order and prove an analogous grafting theorem.

Future directions: Bowers, Casella, Ruffoni, and I are currently working to extend our results to more general non-compact surfaces. One complication in this more general setting is that there is no longer a unique complex structure on the surface, and so any classification scheme will need to take this complex structure into consideration.

[1] Dylan G. L. Allegretti and Tom Bridgeland. The monodromy of meromorphic projective structures. *Trans. Amer. Math. Soc.*, 373(9):6321–6367, 2020.

[2] Shinpei Baba. 2π-grafting and complex projective structures, I. *Geom. Topol.*, 19(6):3233–3287, 2015.

[3] Shinpei Baba. 2π-grafting and complex projective structures with generic holonomy. *Geom. Funct. Anal.*, 27(5):1017–1069, 2017.

[4] Samuel Ballas and Darren D. Long. Constructing thin subgroups commensurable with the figure-eight knot group. *Algebr. Geom. Topol.*, 15(5):3011–3024, 2015.

[5] Samuel Ballas and Ludovic Marquis. Properly convex bending of hyperbolic manifolds. *Groups Geom. Dyn.*, 14(2):653–688, 2020.

[6] Samuel A. Ballas. Finite volume properly convex deformations of the figure-eight knot. *Geom. Dedicata*, 178:49–73, 2015.

[7] Samuel A. Ballas. Constructing convex projective 3-manifolds with generalized cusps. *Journal of the London Mathematical Society*, Nov 2020.

[8] Samuel A. Ballas. Thin subgroups isomorphic to Gromov–Piatetski-Shapiro lattices. *Pacific J. Math.*, 309(2):257–266, 2020.

[9] Samuel A. Ballas, Phil Bowers, Alex Casella, and Lorenzo Ruffoni. Tame and relatively elliptic $\mathbb{CP}^1$-structures on the thrice punctured sphere, 2021.

[10] Samuel A. Ballas and Alex Casella. Gluing equations for real projective structures on 3-manifolds, 2019, 1912.12508.

[11] Samuel A. Ballas and Alex Casella. Neumann–zagier relations for projective gluing equations, in preparation, 2021.

[12] Samuel A. Ballas, Daryl Cooper, and Arielle Leitner. Generalized cusps in real projective manifolds: classification. *J. Topol.*, 13(4):1455–1496, 2020.

[13] Samuel A. Ballas, Daryl Cooper, and Arielle Leitner. The moduli space of marked generalized cusps in real projective manifolds. 2020, 2008.09553.

[14] Samuel A. Ballas, Jeff Danciger, Gye-Seon Lee, and Ludovic Marquis. Properly convex dehn filling, in preparation, 2021.

[15] Samuel A. Ballas, Jeffrey Danciger, and Gye-Seon Lee. Convex projective structures on nonhyperbolic three-manifolds. *Geom. Topol.*, 22(3):1593–1646, 2018.

[16] Samuel A. Ballas and Darren D. Long. Constructing thin subgroups of $\mathrm{SL}(n+1,\mathbb{R})$ via bending. *Algebr. Geom. Topol.*, 20(4):2071–2093, 2020.

[17] Yves Benoist. Convexes divisibles. IV. Structure du bord en dimension 3. *Invent. Math.*, 164(2):249–278, 2006.

[18] Nicolas Bergeron, Elisha Falbel, and Antonin Guilloux. Tetrahedra of flags, volume and homology of SL(3). *Geom. Topol.*, 18(4):1911–1971, 2014.

[19] Martin D. Bobb. Convex projective manifolds with a cusp of any non-diagonalizable type. *J. Lond. Math. Soc. (2)*, 100(1):183–202, 2019.

[20] Jean Bourgain. Some Diophantine applications of the theory of group expansion. In *Thin groups and superstrong approximation*, volume 61 of *Math. Sci. Res. Inst. Publ.*, pages 1–22. Cambridge Univ. Press, Cambridge, 2014.

[21] Jean Bourgain, Alex Gamburd, and Peter Sarnak. Affine linear sieve, expanders, and sum-product. *Invent. Math.*, 179(3):559–644, 2010.

[22] Benjamin A. Burton, Ryan Budney, William Pettersson, et al. Regina: Software for low-dimensional topology. `http://regina-normal.github.io/`, 1999–2021.

[23] D. Cooper, D. D. Long, and S. Tillmann. On convex projective manifolds and cusps. *Adv. Math.*, 277:181–251, 2015.

[24] Daryl Cooper and David Futer. Ubiquitous quasi-Fuchsian surfaces in cusped hyperbolic 3-manifolds. *Geom. Topol.*, 23(1):241–298, 2019.

[25] Daryl Cooper, Darren Long, and Stephan Tillmann. Deforming convex projective manifolds. *Geom. Topol.*, 22(3):1349–1404, 2018.

[26] Daryl Cooper and Stephan Tillmann. On properly convex real-projective manifolds with generalized cusp, 2020, 2009.06569.

[27] Marc Culler, Nathan M. Dunfield, Matthias Goerner, and Jeffrey R. Weeks. SnapPy, a computer program for studying the geometry and topology of 3-manifolds. Available at `http://snappy.computop.org`.

[28] Elena Fuchs and Igor Rivin. Generic thinness in finitely generated subgroups of $\mathrm{SL}_n(\mathbb{Z})$. *Int. Math. Res. Not. IMRN*, (17):5385–5414, 2017.

[29] Daniel Gallo, Michael Kapovich, and Albert Marden. The monodromy groups of Schwarzian equations on closed Riemann surfaces. *Ann. of Math. (2)*, 151(2):625–704, 2000.

[30] Stavros Garoufalidis, Matthias Goerner, and Christian K. Zickert. Gluing equations for $\mathrm{PGL}(n,\mathbb{C})$-representations of 3-manifolds. *Algebr. Geom. Topol.*, 15(1):565–622, 2015.

[31] Stavros Garoufalidis and Christian K. Zickert. The symplectic properties of the $\mathrm{PGL}(n,\mathbb{C})$-gluing equations. *Quantum Topol.*, 7(3):505–551, 2016.

[32] M. Gromov and I. Piatetski-Shapiro. Nonarithmetic groups in Lobachevsky spaces. *Inst. Hautes Études Sci. Publ. Math.*, (66):93–103, 1988.

[33] Subhojoy Gupta and Mahan Mj. Meromorphic projective structures, grafting and the monodromy map. *Adv. Math.*, 383:107673, 2021.

[34] Ursula Hamenstädt. Incompressible surfaces in rank one locally symmetric spaces. *Geom. Funct. Anal.*, 25(3):815–859, 2015.

[35] Jeremy Kahn, Franois Labourie, and Shahar Mozes. Surface groups in uniform lattices of some semi-simple groups, 2020, 1805.10189.

[36] Jeremy Kahn and Vladimir Markovic. Immersing almost geodesic surfaces in a closed hyperbolic three manifold. *Ann. of Math. (2)*, 175(3):1127–1190, 2012.

[37] Jeremy Kahn and Alex Wright. Nearly fuchsian surface subgroups of finite covolume kleinian groups, 2020, 1809.07211.

[38] Dave Witte Morris. *Introduction to arithmetic groups.* Deductive Press, 2015.

[39] Walter D. Neumann. Combinatorics of triangulations and the Chern-Simons invariant for hyperbolic 3-manifolds. In *Topology '90 (Columbus, OH, 1990)*, volume 1 of *Ohio State Univ. Math. Res. Inst. Publ.*, pages 243–271. de Gruyter, Berlin, 1992.

[40] Adam Paszke, Sam Gross, Soumith Chintala, Gregory Chanan, Edward Yang, Zachary DeVito, Zeming Lin, Alban Desmaison, Luca Antiga, and Adam Lerer. Automatic differentiation in pytorch. 2017.